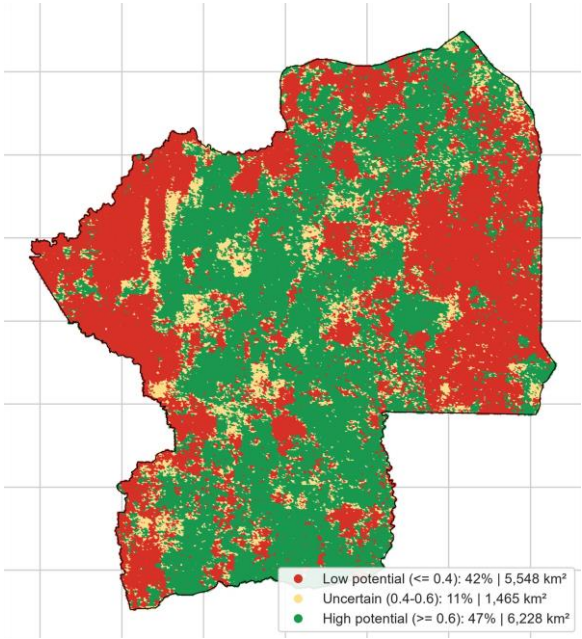
AgroRestore AI

Predicting where forest restoration and agriculture projects are most likely to succeed

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Team NeuralRoots
neuefische Data Science Bootcamp · Capstone Project



Centrale Region, Togo: forest restoration and agriculture potential

Restoration and agriculture money is lost on the wrong sites

Who invests
NGOs and development banks fund reforestation, agroforestry and farmland in West Africa.

How sites are chosen today
Often by expert judgement and accessibility, with little data on how the land has actually behaved.

What it costs
A planting that never regreens or a farm that fails means lost seedlings, labour and loans, and less trust for the next project.

Our question

Which places are most likely to regreen (forest) and suitable for agriculture, and which should funders avoid?

OUR APPROACH · THE DATA SCIENCE LIFE CYCLE

1 Business
Define success: where restoration pays off

2 Data
17 satellite + 2 OpenStreetMap layers

3 Preparation
Sample 99,913 sites, build target, EDA

4 Modelling
4 models, spatial cross-validation

5 Evaluation
Held-out blocks, drivers, opportunity map

6 Deployment
Streamlit app to explore and score sites

Outcome: an opportunity map that sorts every 250 m site into high, uncertain and low potential.

Study area and 20 years of satellite data



13,242
km² in Centrale Region, Togo

99,913
sites on a 250 m grid

2006–2025
climate, vegetation, fire etc.

17
layers via Google Earth Engine

Climate
CHIRPS rainfall, TerraClimate temperature and soil moisture

Soil
iSDAsoil organic carbon and pH, 0–20 cm

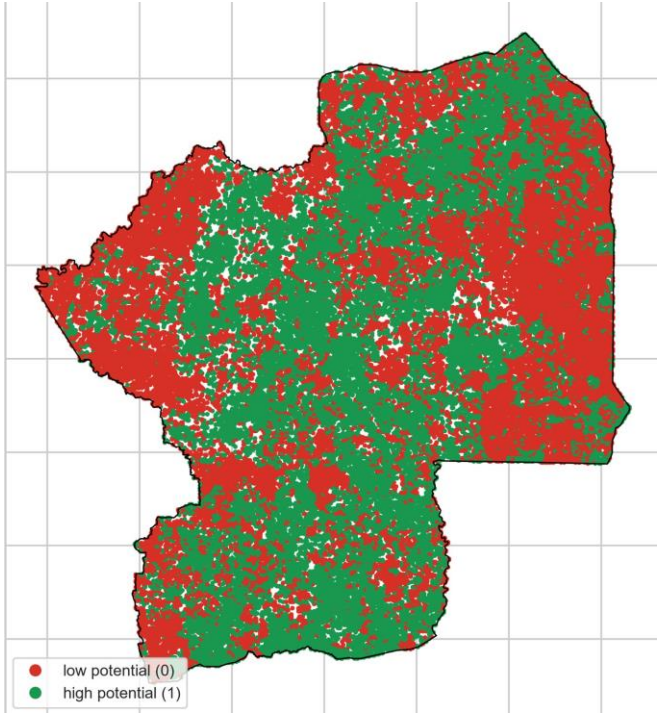
Terrain
SRTM elevation and slope

Land cover
ESA WorldCover built-up and water

Protection & access
WDPA protected areas; OSM roads and settlements

Vegetation & fire
MODIS NDVI and burned area

Turning 20 years of greening into a target



Target: NDVI trend, the slope of annual mean greenness 2006–2025

Low	Dropped	High
bottom 30%	40%, almost no change	top 30%
≤ −0.0007 NDVI/yr		≥ +0.0009 NDVI/yr

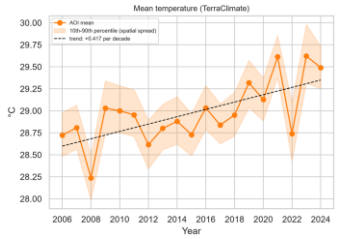
59,948 balanced sites kept for training, with 17 predictors

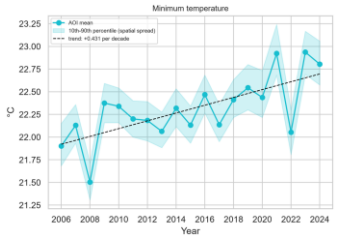
Cleaning: sites missing a critical layer dropped; only layers with more than 95% valid values used

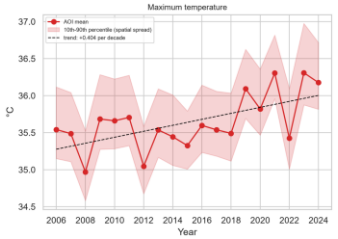
Map: green = regreening (high potential, 1), red = declining (low potential, 0)

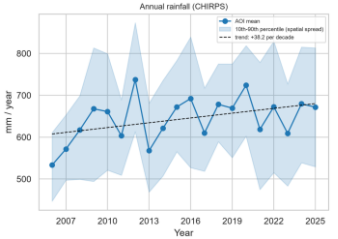
20 years of change: hotter, greener, more fire

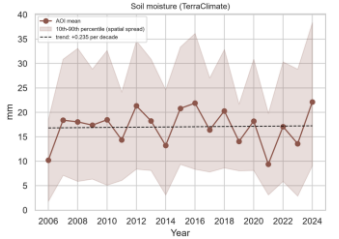
CLIMATE



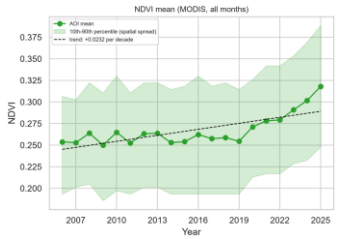


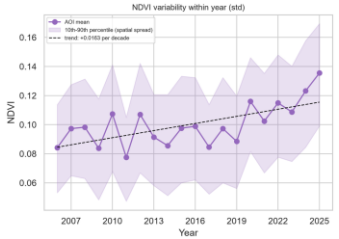


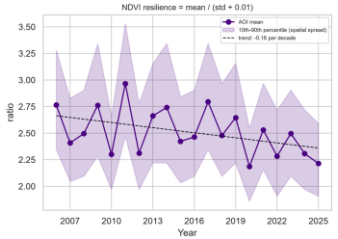


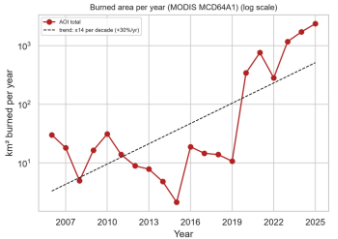


VEGETATION & FIRE









A fast-changing region: a 20-year trend is a better guide than one snapshot.

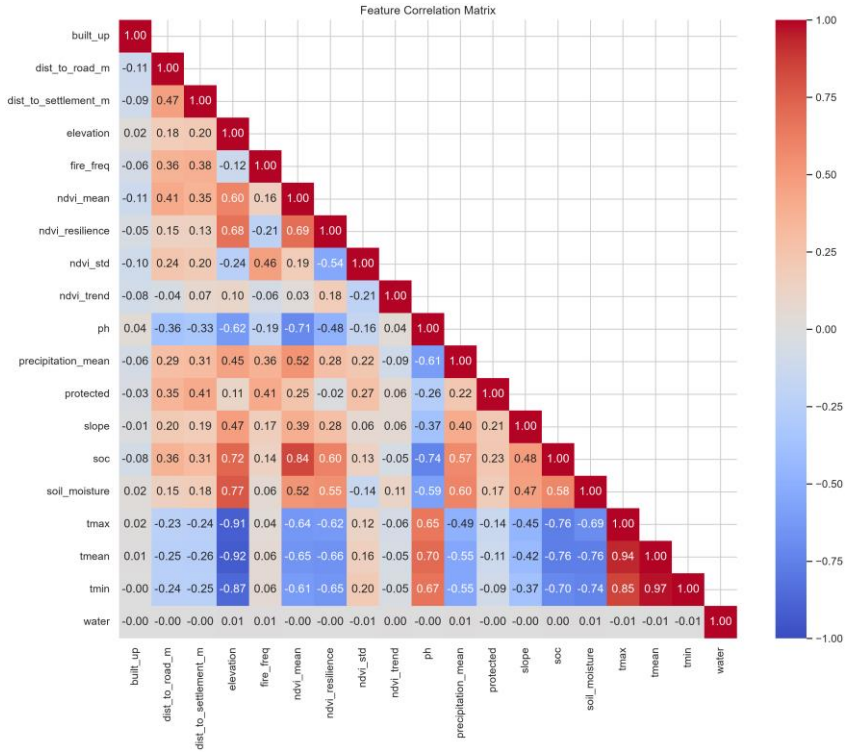
+0.4 °C
per decade, in every temperature

+38 mm
rain per decade, soil moisture flat

Greener since 2019
but less resilient

~2,400 km²
burned in 2025 (was 2–30 a year)

Conditions move together



–0.9
Temperature ↔ elevation
ρ −0.87 to −0.92: higher ground is cooler

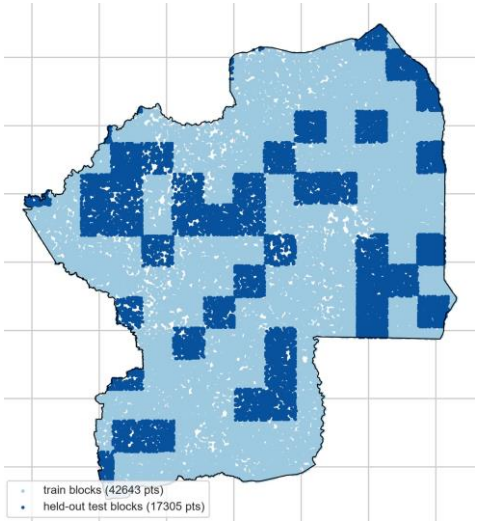
0.84
Soil carbon ↔ greenness
carbon-rich soils carry greener vegetation

0.77
Soil moisture ↔ elevation
wetter soils sit higher up

≤ 0.21
NDVI trend ↔ any one layer
no single layer gives the answer away

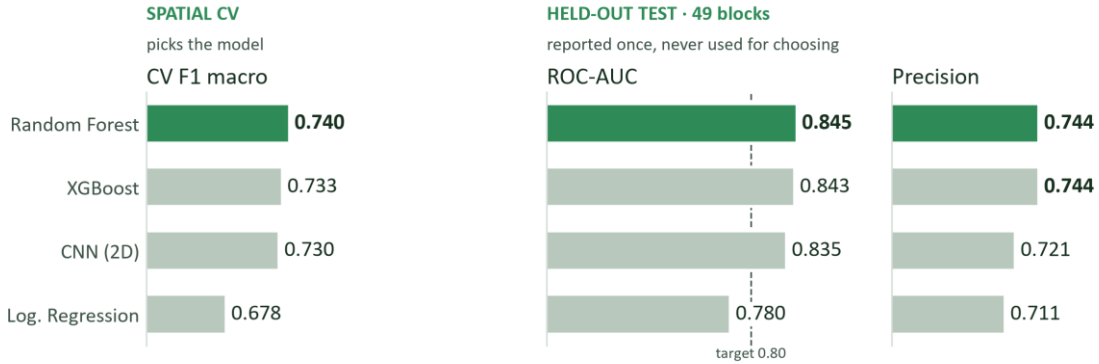
Tree models cope with correlated inputs, but one driver can stand in for another: read drivers as groups (water, terrain, temperature).

Spatial validation: Random Forest selected



10 km blocks: 49 held out as the final test (17,305 sites); 114 for training (42,643 sites).

Four models: Random Forest, XGBoost, Logistic Regression, CNN on 5×5 patches.



Why not a random split?

Neighbouring sites look alike, so a random split leaks answers and inflates scores.

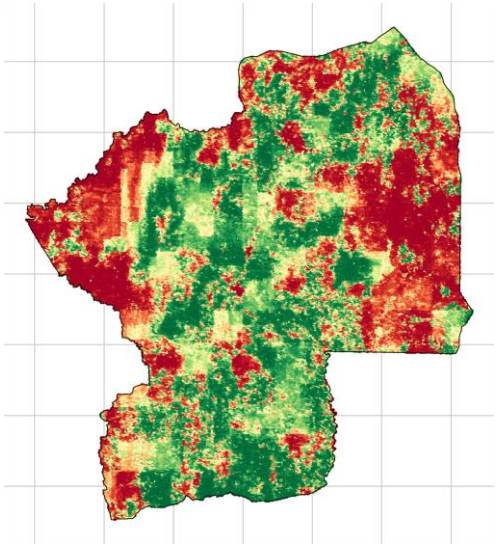
Tuning and selection

Grouped 5-fold cross-validation on the training blocks, scored on F1 macro. Random Forest highest at 0.740.

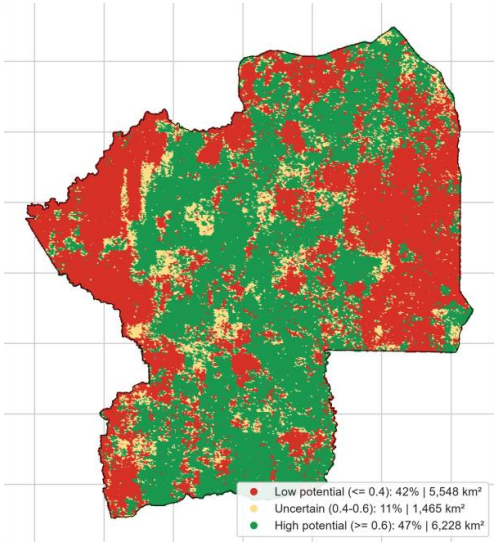
Fair test and targets

Test set used once, never for choosing. Target ROC-AUC ≥ 0.80 met by RF, XGBoost and CNN. Precision: share of flagged sites that really regreened.

The opportunity map and what drives it



Probability of high potential



Tiers at 0.40 and 0.60

Random Forest, 99,913 sites. *Green belt through the centre; low potential in the far west and east.*

High P ≥ 0.60

Uncertain 0.40 – 0.60

Low P ≤ 0.40

WHAT DRIVES RESTORATION POTENTIAL

#1 **Soil moisture**
Rainy-season water in the soil (Jul–Oct): the strongest signal

#2 **Vegetation stability**
NDVI variability and resilience of the starting vegetation

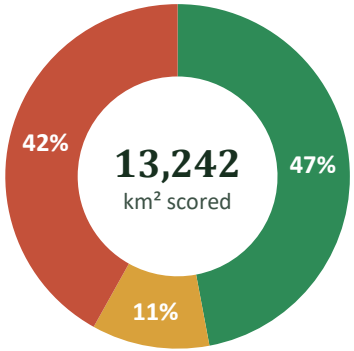
#3 **Elevation**
Terrain, and the temperature gradient that comes with it

#4 **Min. temperature**
Night-time heat that stresses young plants

How we know: permutation importance on the held-out blocks. RF and XGBoost agree on the top two; road and settlement distance rank mid-table.

47% rated high; 34% where all four agree

RANDOM FOREST TIERS · 13,242 KM² SCORED

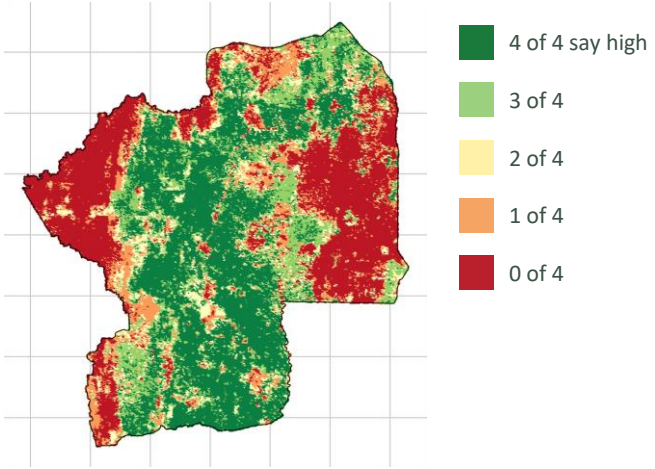


6,228 km² 47% of region
High potential
Prioritise for restoration and agroforestry investment

1,465 km² 11% of region
Uncertain
Verify in the field before committing funds

5,548 km² 42% of region
Low potential
Avoid, or plan different interventions (soil and water first)

WHERE ALL FOUR MODELS AGREE



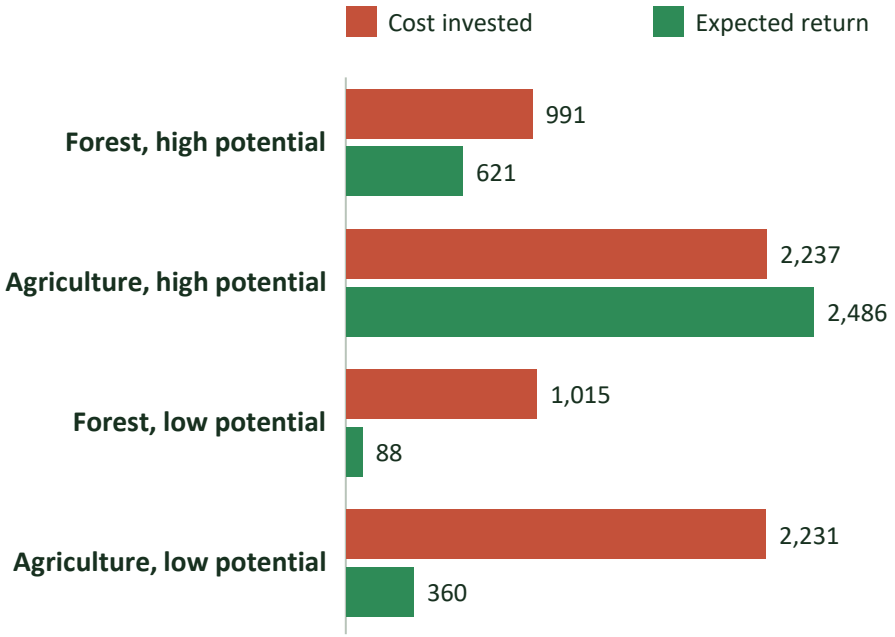
Random Forest alone rates 47% of sites high. Requiring **all four models** to agree narrows it to 34%: fewer sites, more confidence.

34%
Priority zones: all four models say high

27%
Avoid zones: all four models say low

Only high-potential land pays back

USD per hectare over 20 years, in today's money



+249
USD/ha NPV

Agriculture, high potential
ROI +11%; pays back on 70% of these sites

−370
USD/ha NPV

Forest, high potential
Earns 89 USD/ha/yr on average vs 346 for crops (142 once in full production); breaks even at ≈ 226

< 0
NPV, both uses

Low potential
ROI −84% to −91%: avoid, or fix soil and water first

NPV = expected return − cost. **ROI** = NPV ÷ cost. Expected return = revenue × P(high) × fire risk. Forest income = carbon credits (8.9 tCO_2e/ha/yr, 80% sold, 15.6 USD/t) + fuelwood (year 6+) + shea (year 15+). Inputs: FAO, IPCC, World Bank, WFP, Ecosystem Marketplace.

From notebook to decision tool: Streamlit app

SEVEN APP PAGES

- **Overview** · Question, data, method, key scores and tier areas
- **Opportunity map** · Per-model map, adjustable thresholds, CSV export
- **Explore the data** · Conditions per tier, scores per variable, correlations
- **Model performance** · Results, targets, tuning gain, feature importance
- **Cost and return** · Cost, expected return, NPV and ROI per hectare for high and low potential
- **Score a site** · Pick a site, change its conditions, get scores from all models
- **Score a new region** · Type a place name: Earth Engine data, four model scores, range warnings

HOW AN NGO OR BANK WOULD USE IT

1. Screen
Filter the map to high-potential sites where all four models agree

2. Shortlist
Remove protected areas, rank by NPV and ROI, export as CSV

3. Verify
Send field teams first to the shortlist and to uncertain sites

4. Invest & monitor
Fund the verified sites and track NDVI after planting

The goal: put money and field visits where the land has the best chance, and steer it away from the 27% of sites every model rates low, to cut the cost of failed projects.

Thank you

Questions? Live demo of the Streamlit app on request.

SCAN TO CONNECT

Project on GitHub

Kueshi Semanou Dahan on LinkedIn

NEXT STEPS

Field validation 01
Test the predictions against restoration project records and field surveys

More regions 02
Train across several regions of Togo and West Africa so new areas score reliably

Local cost data 03
Use real project budgets; add timber, fruit and biodiversity income

Kueshi Semanou Dahan · NeuralRoots · AgroRestore AI